

Exam Date & Time: 12-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: ECE 2123- SIGNALS & SYSTEMS

SIGNALS and SYSTEMS [ECE 2123]

Marks: 50

Duration: 180 mins.

A

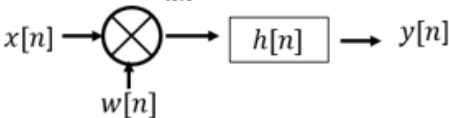
Answer all the questions.

Missing data may be suitably assumed.

- 1A) Sketch the signal $x(t) = u(t + 1) + 2u(t) - u(t - 3) - 2u(t - 5)$. Find and sketch the even and odd part of the signal. (4)
- 1B) Let $x(t) = u(t) - 2u(t - 1) + u(t - 2)$. Sketch $x(3 - 3t)$ (3)
- 1C) Classify the following signals into energy, power or neither of these. Also calculate energy and power of these signals. (3)
- i) $e^{3t}u(-t)$, ii) $e^{-2|t|}$
- 2A) Estimate the step response of the LTI system with impulse response (4)
- $$h(t) = \begin{cases} e^{2t}, & t < 0 \\ e^{-3t}, & t > 0 \end{cases}$$
- 2B) A typical multipath channel has direct path and a reflection path with delay 2 units. The model is then $y[n] = x[n] - a x[n - 2]$ where $x[n]$ is transmitted signal and $y[n]$ is received signal. Estimate the impulse response of a causal inverse system to recover $x[n]$ from $y[n]$. Is such inverse system practically stable? (3)
- 2C) Sketch the DF I and DF II implementations for the following systems described by difference equation (3)
- $$2y[n] + y[n - 1] - 2y[n - 3] = 6x[n - 1] + 4x[n - 2]$$
- 3A) Estimate the convolution of the following signals using Fourier Transform (4)
- $$x_1(t) = te^{-t}u(t)$$
- $$x_2(t) = te^{-2t}u(t)$$
- 3B) Determine the DTFS coefficient of the following signal using method of inspection (3)
- $$x[n] = \cos\left(\frac{8\pi}{19}n + \frac{\pi}{4}\right)$$
- 3C) Determine the Fourier Transforms of the following signal using tables of transforms and properties (3)
- $$x(t) = \cos(2\pi t)e^{-t}u(t)$$
- 4A) Determine the difference equation description for the systems with the following impulse response using Fourier representation (4)
- $$h[n] = \delta[n] + 2\left(\frac{1}{2}\right)^n u[n] + \left(\frac{-1}{2}\right)^n u[n]$$

- 4B) Consider the discrete time system depicted in the figure 4B.1.

Let $h[n] = \frac{\sin(\frac{\pi}{2}n)}{\pi n}$. Determine $y[n]$ if $x[n] = \frac{\sin(\frac{\pi}{4}n)}{\pi n}$, $w[n] = (-1)^n$



(3)

Figure 4B.1

- 4C) Consider $x(t) = \left(\frac{1}{800\pi}\right)x_1(t)x_2(t)$, where $x_1(t) = \frac{\sin(400\pi t)}{\pi t}$ and $x_2(t) = \frac{\sin(600\pi t)}{\pi t}$. Determine the Nyquist rate. Plot the spectrum of sampled signal at Nyquist rate of sampling.
- (3)

- 5A) Determine the transfer function, difference equation representation of the system with the impulse response $h[n] = (1/3)^n u[n] + (\frac{1}{2})^{n-2} u[n-1]$.
- (4)

- 5B) Use the method of partial fraction expansion to obtain the time domain signal for the function
- $$X(z) = \frac{z^2 - 3z}{z^2 + \frac{3}{2}z - 1}, \quad \frac{1}{2} < |z| < 2.$$
- (3)

- 5C) Determine the time domain signal for the function $X(s) = \frac{4s^2 + 8s + 10}{(s+2)(s^2 + 2s + 5)}$ with the ROC
- (i) $\text{Re}(s) < -2$
 - (ii) $\text{Re}(s) > -1$
 - (iii) $-2 < \text{Re}(s) < -1$
- (3)

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